

Multi-Stage Solicitation

Chained forecast elicitation: representations, and the designs the theory supports

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Abstract

Elicitation mechanisms chain when one mechanism's output is the next one's message or settled outcome. What can be chained, and how to score it, turns on what a mechanism emits: a *point* (a price or a forecast) or a *distribution* (a density, a percentile, or a rank). This note sorts the chains by representation, marks which the theory supports, and then reviews which have been built. Point outputs chain as ordinary derivatives. Distributions chain more richly, provided an exogenous outcome anchors the chain: a residual stage is then scored on the part its predecessor left uniform, and a downstream contribution may be judged in its own market or, for the log score equivalently, converted into a top-level forecast and judged there. Without an anchor the chain is indeterminate. Only then do we turn to practice: the microprediction platform's percentile and copula streams, monteprediction, and the residual contests at CrunchDAO.

1 Introduction

Deployed forecasting contests almost always run one level of competition against one internal model: Numerai pays each staked submission its marginal contribution to a single meta-model (Craib et al. 2017), CrunchDAO blends a contest into one ensemble, the IARPA prediction polls aggregated by track record (Atanasov et al. 2017). The alternative is to *chain*: let one mechanism's output be the message, or the settled outcome, of the next, so that calibration and dependence become the subject of their own games.

Whether two mechanisms chain turns on what the upstream one emits, and that is the organizing question. A stage emits either a point or a distribution, and sorting the four ways one can feed the next leaves only one combination worth much (§2). The rest follows from there. Section 3 gives the operator that chains and the two ways to score a stage inside a chain. Section 4 draws the line between the designs the theory backs and the one it does not. Section 5 is what has been built. The single-stage dictionary the chains draw on, proper scores and market makers and pools as one convex object, is a companion paper (Cotton 2026a); here the object is the chain.

2 Two representations, and the chains between them

A stage of a chain emits one of two things.

A **point output** is a scalar summary: a price, a point forecast, a single reported percentile. A **probabilistic output** is a full object over the outcome space: a density or distribution function,

the probability-integral transform (rank) of an outcome, or a finite cloud of samples standing in for a density. The distinction is not cosmetic: it decides what a downstream stage can be paid to get right.

Write the upstream stage's output as the message or the settlement transform of the downstream stage. Four chains are possible, and only some carry content.

- **Point $\square$ point.** A second market settles on the first's price: every derivative, the electricity virtual bids and transmission rights written on a day-ahead price, a market on another market's reported number. This is ordinary derivative structure, a translation of a point output; it is ubiquitous and is not composition of *elicitation* — no distribution is elicited about the upstream object.
- **Probabilistic $\square$ probabilistic.** The upstream distribution, percentile, or rank is the downstream stage's message or settled object. This is the case with content, and the rest of the note is about it. It needs one thing to be well posed: an exogenous outcome somewhere in the chain to anchor the scores (§4).
- **Point $\square$ probabilistic.** A point output carries no distribution, so a downstream distributional stage has nothing to elicit *about it* unless a distribution is supplied. The natural way to supply one is to model the point forecast's error: a stage that emits a point leaves a residual, and a second stage elicits the residual's law. Point-to-probabilistic chaining is the residual construction of §3.
- **Probabilistic $\square$ point.** A distribution can be summarized by a number and a point market chained on the summary, dropping whatever the summary omits. How much that is depends on how many summaries you chain: one quantile or moment loses most of the shape, but a battery of point markets, one per moment or one per quantile on a grid, keeps proportionately more, and reassembling a distribution from them is the market analogue of method-of-moments estimation. In the limit a full quantile function is the distribution again. So this is a lossy encoding of the probabilistic-to-probabilistic case, as coarse or as fine as the number of summaries, rather than a design of its own.

So the representation settles what chains to what. Point outputs chain among themselves as derivatives, distributions among themselves as elicitation, and the two worlds meet in two ways: a residual stage adds a distribution to a point by scoring its error (§3), and a summary, or a battery of them, drops a distribution back to points.

3 Composing a chain, and the two ways to score it

A *stage* is a transducer (Mealy machine) over one message type. Let Dist be the set of distributional beliefs; a stage carries a wealth state w , consumes participant reports in Dist^m together with, where it settles, a realized outcome x , and emits an aggregate belief and a transfer vector,

$$M : (\text{Dist}^m, w, x) \mapsto (\text{Dist}, w', \pi).$$

A scoring rule is the transfer component of a stage; a market maker is a stage whose state is the inventory vector and whose emitted belief is the price; an opinion pool is a stage with no outcome argument and zero transfers; a parimutuel is a stage whose transfers are a pot split. The operators that build and combine stages (Sequentialise, Pool, Ensemble, Merge, Conjugate)

and the convex dictionary behind them are the companion algebra (Cotton 2026a). One operator does the chaining.

Residual. Let a stage emit the aggregate F_1 for an outcome Y , and let a second market elicit a distribution for the residual $U = F_1(Y)$, settling at the realised $u = F_1(y)$. If F_1 is the true conditional law then U is uniform (the probability integral transform (Cotton 2026a, Prop. 4)) and the second market has nothing to price; whatever structure remains in the residual is the second stage's edge.

Proposition 1 (the correction is a multiplicative reweighting). *Let F_1 be strictly increasing onto $(0, 1)$ with density $p_1 > 0$ and let the residual market's consensus be a distribution H on $[0, 1]$ with $H(0) = 0$, $H(1) = 1$ and density g . The composed forecast $F = H \circ F_1$ has density*

$$p(y) = p_1(y) g(F_1(y)),$$

so $\log p(y) = \log p_1(y) + \log g(u)$ with $u = F_1(y)$: the chain's log score is the sum of stage log scores, and the residual stage is paid by a proper score on u alone.

Proof. Chain rule: $F'(y) = g(F_1(y)) p_1(y)$; take logarithms. The residual score $\log g(u)$ is the logarithmic score (Cotton 2026a, Thm 1) applied to the report g and outcome u , hence strictly proper for the law of U . ■

Local scoring versus top-level conversion. A contribution to the second market can be judged two ways. *Locally*, the residual stage is scored on its own object u , by $\log g(u)$. *At the top level*, the contribution is converted into a forecast of Y — the composed density $p_1(y) g(F_1(y))$ — and scored against the ultimate outcome by $\log p(y)$. Proposition 1 says these coincide: $\log p(y) = \log p_1(y) + \log g(u)$, so up to the constant $\log p_1(y)$ that no downstream report can move, the two scores rank downstream contributions identically. This is the choice raised in the stacked-lottery design (Cotton 2020b, slides 29-31): run the secondary market on its own residual, or fold every secondary contribution up to the top level and score it there. For the log score they are the same mechanism; the equivalence is special to scores that are additive under composition, and fails for a general proper score.

Multiplying the density by a ratio fitted to what the current model gets wrong is the functional-gradient step of boosting under log loss (Mason et al. 1999; Friedman 2001), so a chain of residual markets is stagewise boosting with wealth as the learning rate.

Spec. Serialise a pipeline to data and search over it; the mechanism analogue is a market over pipelines. Open.

Stagewise play. Call a profile of reports a *stagewise equilibrium* if no participant gains by a deviation confined to a single stage, all other stages' reports held fixed *and the inputs and settlement transforms of every other stage clamped at their pre-deviation values*. This is the pipeline version of the myopic-trader assumption standard in the market-scoring-rule literature (Hanson 2007; Chen and Vaughan 2010). The clamp is not cosmetic: a stage's output is wired into later settlement transforms (the residual point $u = F_1(y)$, or a rank vector in a copula stage), so an upstream deviation moves downstream payoffs even when every downstream report is held fixed.

Proposition 2 (single-stage guarantees compose under clamped stagewise play). *Suppose each stage of a pipeline, taken in isolation with its inputs and settlement transform fixed, makes the truthful report a best response: strict propriety for externally funded scoring stages (Cotton 2026a, Thm 1), the sequential scoring (Cotton 2026a, Thm 2) for market stages, the price-taking pot-split analysis of the point-cloud paper (Cotton 2026c) for parimutuel stages, and the residual score of Proposition 1 for correction stages. Then truthful reporting at every stage is a stagewise equilibrium of the pipeline. If*

moreover no participant reports to a stage upstream of one in which they hold a position (disjoint stage membership suffices), the clamp is vacuous for every feasible deviation, and truthful reporting survives unrestricted single-stage deviations.

Proof. With the other stages' inputs and transforms clamped, a deviation confined to stage k changes the deviator's payoff only through stage k 's transfer map, and the stage- k hypothesis makes the truthful report a best response. For the second claim: a stage- k deviation moves stage k 's transfer and, through stage k 's output, transfers strictly downstream of k ; a deviator with no downstream position collects none of the latter, so the propagation is payoff-irrelevant to them. ■

Proposition 2 is not a Nash equilibrium of the composed game. When the same participant reports upstream and holds exposure downstream, the deviation propagates through the settlement transform, and a downstream stake is a derivative written on an upstream settlement, with the attendant incentives to distort the underlying (Kumar and Seppi 1992; Jarrow 1994; Hanson and Oprea 2009; Ostrovsky 2012). Disjoint membership, zero downstream exposure for upstream reporters, or exogenous freezing of the settlement transforms restore the proposition; the dynamic game without them is outside this note's scope.

4 Which designs the theory supports

The propriety of the chain rests on one requirement: an exogenous outcome anchors it. Every score above is paid against a realised y or its rank u ; strip the outcome out and nothing pins the reports down. This is the line between the designs the theory supports and the one it does not.

Anchored probabilistic chains (supported). A pipeline whose final stage settles on a real outcome, with proper local scores at each stage, is well founded: Proposition 2 makes truthful reporting a stagewise equilibrium, and Proposition 1's additivity makes local scoring and top-level conversion the same mechanism. Residual chains (boosting), and the copula factoring below, are of this kind.

Unanchored chains (not supported). A chain in which one market settles on another market's displayed probability, with no exogenous outcome anywhere, has no scoring anchor: the equilibrium is indeterminate and the mechanism runs as a beauty contest. Manifold's resolves-to-market markets are the live example; they are play-money precisely because there is nothing to be proper about.

Samples as the message (supported, and what contests collect). The message type has so far been a distribution given exactly. Deployed contests collect something rougher: a finite cloud of samples, smoothed into a density before settlement. The companion point-cloud paper gives the elicitation result the chain needs (Cotton 2026c):

Proposition 3 (sample-based elicitation; Cotton (2026c), Thms 1-2). *Score the bandwidth- h kernel density estimate of a submitted cloud by the logarithmic score. Settled at the raw outcome, the optimal cloud is drawn from a deconvolution of the belief when one exists (for Gaussian beliefs and kernels with belief variance exceeding h^2 , the belief with h^2 removed from the variance). Settled at the outcome jittered by the same kernel, truthful sampling is optimal, and strictly so whenever the kernel's characteristic function is nonvanishing on a dense set: the smoothing channel is injective on laws.*

With jittered settlement the cloud is a valid message, and the operators act on clouds point-wise: Conjugate pushes each sample through the map, a linear Pool is a wealth-weighted union of clouds, Residual ranks each sample through the upstream CDF. One caveat governs the order of operations — smoothing and a nonlinear map do not commute, so the kernel and its jitter

belong in the settlement coordinates, immediately before settlement and nowhere else.

Dependence by rank (supported). A multivariate settlement factors. By Sklar’s theorem (Sklar 1959) a joint law with continuous margins $F_1, \dots, F_d$ and copula C has density $p(x) = \prod_i f_i(x_i) \cdot c(F_1(x_1), \dots, F_d(x_d))$.

Proposition 4 (the log score factors). *For a joint report assembled from margin reports f_i and a rank-stage report c , a density on $[0, 1]^d$,*

$$\log p(x) = \sum_{i=1}^d \log f_i(x_i) + \log c(u), \quad u_i = F_i(x_i),$$

so a pipeline of d margin stages and one rank stage settled on the rank vector u pays every stage a logarithmic score of its own object, and the chain’s log score is the sum. Given the margin reports, the rank stage’s score is strictly proper for the law of the rank vector, which has uniform margins and is the copula of the joint iff the margin reports are correct.

Proof. Take logarithms in Sklar’s factorization; the summands are the stage scores. Strict propriety of the rank stage is the logarithmic score (Cotton 2026a, Thm 1) on densities over $[0, 1]^d$, the transform $x \mapsto u$ fixed once the margins are given. ■

This is the same additivity as Proposition 1, one dimension per margin: the rank stage is the multivariate Residual operator, eliciting dependence separately from level, and again local scoring of the rank stage equals its top-level conversion. The construction is the two-stage estimation logic of inference functions for margins (Joe 1997) read as a market design.

A worked anchored chain: betting against a conformal predictor. A point predictor leaves a residual R ; conformalization re-levels its marginal law but prices the residual flat in the input. An entrant who conditions on the input X collects what the flat predictor discards as bankroll growth, at the rate $I(R; X)$, the mutual information between residual and input, while the band’s marginal coverage stays exact; marginal coverage is the break-even statement, the conditional information is the rent. The full account, with the Gaussian rate $-\frac{1}{2} \log(1 - \rho^2)$ and the anytime-valid measurement, is a standalone note (Cotton 2026b). It is a point-to-probabilistic residual chain with the outcome as anchor, so the theory above applies whole.

5 What has been built

The supported designs are rare in practice; one platform ran several at once.

The microprediction platform. The base game was a pool on a single live number. Anyone could open a stream for a quantity they cared about; contributors submitted a fixed-size bundle of Monte Carlo scenarios, 225 of them, and when the number arrived the pot was split by how near the scenarios fell. Paying by log-wealth made a contributor’s best move to submit scenarios that matched her honest distribution (Cotton 2022). That is the nearest-the-pin rule, and it only ever settled one scalar.

To make calibration and dependence into games of their own, the platform turned each of them into another scalar. Calibration first. Once the community has forecast a quantity, ask where the outcome actually landed in the forecast distribution: its percentile, or the z-score you get by pushing that percentile through a normal. Forecast well and those z-scores look standard normal; forecast badly and they drift or spread out. A z-score is just another number, so it opened its own stream, $z1\sim$, and predicting $z1\sim$ meant predicting the first game’s miscalibration. In the language of §3 this is the probability-integral transform run as a residual stage.

Dependence used the same idea in reverse. The joint behaviour of two streams is two-dimensional, but a pool settles one number, so the two community percentiles were folded into a single number by the Morton z-curve: interleave the binary digits of the two coordinates, the way a geohash packs latitude and longitude into one string. A pool on the folded number was a market on the copula of the pair, $z2\sim$; three streams folded the same way gave $z3\sim$. A 2020 copula contest ran exactly this on the five-minute comovements of five cryptocurrencies, contributors submitting 225 samples packed through the z-curve (Cotton 2020a). This is the rank factoring of Proposition 4, in production. A related stacked-lottery design let competing algorithms contribute monotone calibration maps that composed into one forecast (Cotton 2020b, slides 29-31). The platform is retired.

Its successors. monteprediction is the base game with one stage repeated rather than chained. Each weekly submission is about a million joint scenarios of eleven sector-ETF returns, and the pot is split in proportion to the density a participant places on the realised vector (Cotton 2024c, 2024a); wealth threads across rounds, and the contest has run since January 2024. The MidOne contests at CrunchDAO priced residual densities directly (CrunchDAO 2024; Cotton 2024b), a residual stage without the sample smoothing.

For contrast, the point-output world. Chaining on point outputs is everywhere: derivatives, the electricity virtual bids and transmission rights on day-ahead prices (Jha and Wolak 2023), a market on another market's reported number. But it is derivative structure, not elicitation composition. And the racetrack's win versus exotic pools, index versus single-name option books, and tranche versus single-name CDS run margins and dependence in *parallel* books that settle independently, consistency left to arbitrage (Harville 1973; Hausch et al. 1981); parallel is not chained, and the books can disagree.

6 Open problems

1. *Conservation of edge.* Wealth conservation is automatic, a sum of stage-level zeros, and is not the invariant. The invariant that can fail is the edge a truthful participant holds over the price, $D_G(q, \pi)$. For the logarithmic score the edge is $KL(q\|\pi)$ and the data-processing inequality settles it: an interface channel cannot increase it, and preserves it exactly when the channel is sufficient for the pair (Csiszár 1967). No such inequality holds for a general Bregman edge. For the quadratic generator, merging the first two of three outcomes sends $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and $q = (\frac{13}{30}, \frac{13}{30}, \frac{2}{15})$, with $D_G(q, \pi) = \frac{3}{50}$, to a pair with $D_G(qK, \pi K) = \frac{2}{25}$: coarsening increased the Brier edge. Open: characterize the generators and channels for which $D_G(qK, \pi K) \leq D_G(q, \pi)$.
2. *Residual markets.* The single-step correction is a multiplicative reweighting of the upstream density (Proposition 1); a chain of residual markets is stagewise boosting with wealth as the learning rate. Open: the microstructure (who funds each residual pot, and when it settles relative to the stage before), whether a forecaster free to enter several stages prefers withholding information upstream to sell it downstream, and whether the iterated correction converges to the true conditional law as boosting does. Split-conformal prediction is the one-participant degenerate case, with its rent $I(R; X)$ as the value left on the table (Cotton 2026b).
3. *Copula markets.* A rank-settled stage elicits dependence separately from margins (Proposition 4). Open: the equilibrium when the same participants trade margin and copula stages, and the choice of embedding for $d \geq 2$, the space-filling curves as deployed against the

random one-dimensional projections of the point-cloud paper (Cotton 2026c).

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